Notebook PDF export

Kernel: Python 3 • Code cells: 1 • Markdown cells: 0

In [1]:

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_moons
from sklearn.cluster import DBSCAN
# Generate sample data
X, _ = make_moons(n_samples=200, noise=0.05, random_state=42)
# Apply DBSCAN
dbscan = DBSCAN(eps=0.2, min_samples=5)
labels = dbscan.fit_predict(X)
# Plot results
plt.figure(figsize=(8, 5))
plt.scatter(X[:, 0], X[:, 1], c=labels, cmap='viridis')
plt.title('DBSCAN Clustering')
plt.xlabel('Feature 1')
plt.ylabel('Feature 2')
plt.show()
# Summary
n_clusters = len(set(labels)) - (1 if -1 in labels else 0)
n_noise = list(labels).count(-1)
print(f"Clusters: {n_clusters}, Noise points: {n_noise}")
```

Output:

```
Clusters: 2, Noise points: 0
```